

Investments in Human Capital in the Age of AI

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Abstract

How does Artificial Intelligence reshape human capital investment decisions? I develop a task-based model where risk-averse agents make irreversible training choices based on AI's expected impact and uncertainty. Calibrated to the US labor market, the framework decomposes AI shocks into expected automation, expected augmentation, and their associated uncertainties. The model yields three key results. First, expected automation drives a structural reallocation, causing workers to systematically avoid training paths facing high automation risk, while choosing paths requiring more manual or social skills. Second, direct AI impact and labor scarcity balance each other; as a result, expected wages rise across all training paths by approximately +14%. Third, automation uncertainty generates an endogenous risk premium of up to +2.3%.

1 Introduction

For most of the twentieth century, technological progress has pointed in the direction of incentivizing investments in human capital. Successive waves of automation primarily displaced routine, mechanical tasks concentrated in low and middle-skill occupations, while simultaneously raising the demand for high-skill workers with higher education (Acemoglu and Autor, 2011; Autor et al., 2003). In this environment, a college degree has been a relatively safe investment, offering high expected earnings and low unemployment risk. The secular rise in the college wage premium and in educational attainment throughout the latter half of the twentieth century is largely a product of this dynamic (Autor, 2019).

Artificial Intelligence (AI) fundamentally disrupts this long-standing relationship. Unlike previous technologies, generative AI demonstrates a high degree of proficiency precisely in the non-routine, cognitive tasks that have historically been the exclusive domain of high-skill workers (Eloundou et al., 2024; Tomlinson et al., 2025; Massenkoff and McCrory, 2026; Webb, 2019). The implications are potentially far-reaching: the

very occupations that attracted generations of students to invest heavily in human capital are now among the most exposed to this new technology.

Importantly, AI’s impact on training choices also has important implications for expected wages. If individuals systematically avoid training paths that are highly exposed to AI, this creates scarcity of workers with that training, which mechanically pushes up their expected wages. Conversely, the inflow of workers into safer training paths creates a congestion of workers that puts downward pressure on wages in those training paths. In this paper, I show that this mechanism has the potential to counteract the direct effects of AI: training paths that are expected to be negatively impacted by AI lose workers, and this scarcity pushes up their expected wages, while training paths that are expected to be positively impacted by AI attract workers, and this congestion pushes down their expected wages. The two forces balance each other, and the final outcome is that expected wages increase for all training paths by approximately +14%.

Another critical feature of this technological breakthrough, however, is its profound uncertainty. The net effect of AI on labor demand is not only unknown, but its expected impact and the magnitude of the uncertainty vary dramatically across occupations and training paths. It is very difficult to predict exactly which tasks will be automated and which will be augmented. Hence, risk-averse individuals may systematically avoid training paths that are relatively uncertain, irrespective of their expected returns. This flight from uncertainty operates through the same scarcity channel: it reduces labor supply in risky paths and generates an endogenous risk premium that pushes their expected wages upward.

To formalize and quantify these dynamics, I develop a task-based model in which heterogeneous, risk-averse agents choose the optimal irreversible human capital investment among 13 training paths to maximize their expected lifetime utility, net of idiosyncratic investment costs. The key innovation is the decomposition of the AI shock for each training path j into four distinct components: an expected automation effect $\mu_{A,j}$, an expected augmentation effect $\mu_{\gamma,j}$, and their respective variance components $\sigma_{A,j}^2$ and $\sigma_{\gamma,j}^2$, which capture the uncertainty about AI’s ultimate capabilities. The individual discrete choice problem aggregates to a tractable multinomial logit equilibrium.

I calibrate the model to the US labor market. The task-level AI shocks are derived from Althoff and Reichardt (2026), who computed for each task whether it is theoretically automatable with generative AI, as well as the expected augmentation factor if generative AI instead acts as an augmentation technology in that task. Moreover, I rely on the current AI task penetration (Massenkoff and McCrory, 2026), which measures the current AI usage in each task according to Anthropic data. The employment shares across 13 training paths are constructed from the IPUMS American Community Survey (ACS) for 2022, while the median wages are collected from the Bureau of Labor Statis-

tics (BLS). I recover the structural training cost parameters by inverting the pre-AI steady-state equilibrium (2022), and simulate the resulting post-AI equilibrium.

The model delivers three main results, linking theoretical mechanisms to their quantitative implications for the US economy. First, expected automation is the primary driver of labor reallocation, shifting workers away from highly exposed training paths (legal professions, computer science, and business) toward safer alternatives (technical manual jobs and education) or to no training at all. Also, uncertainty about automation amplifies this phenomenon: part of the reallocation is driven by pure risk aversion.

Second, the model predicts an across-the-board increase in expected wages by approximately 14%. Training paths very exposed to automation are chosen less (up to -16%); as a result, the scarcity of workers in those training paths mechanically pushes up their expected wages and acts as a countervailing force to the direct negative effect of AI.

Third, when we add uncertainty, it uniquely generates an endogenous risk premium (up to 2.3%). This is a pure general equilibrium effect rather than the standard risk premium arising from investment in a risky asset.

These three mechanisms interact in ways that are difficult to detect using reduced-form methods alone. A highly exposed training path may see its expected wage rise purely due to the labor scarcity, even if AI has a negative direct expected impact. Conversely, a safer path may experience disappointing wage growth due to labor market congestion. This work provides a structural framework to disentangle these intertwined effects.

This paper contributes to three strands of the literature. First, it builds on the large empirical literature on AI’s impact on the labor market (among many others, Acemoglu and Restrepo 2018; Autor and Thompson 2025; Agrawal et al. 2025; Ide and Talamàs 2025; Acemoglu et al. 2025; Gans and Goldfarb 2026), by providing a structural model that allows for a clean decomposition of the supply-side response to the future expected demand shift and uncertainty related to AI. Second, it contributes to the literature on human capital investment under uncertainty (Levhari and Weiss, 1974; Cunha et al., 2005; Cunha and Heckman, 2016), by introducing a new form of risk in the context of human capital investment decisions. Third, it connects to the macro literature on skill-biased technological change (Acemoglu and Autor, 2011; Autor et al., 2003; Autor, 2019) by characterizing the novel mechanism through which AI reshapes economic incentives in ways that are qualitatively different from prior automation waves.

In this regard, the closest paper to this work is Althoff and Reichardt (2026), which provides the foundational task-level framework for how AI simultaneously automates, augments, and simplifies human labor. While they embed these mechanisms into a rich dynamic model of occupational choice, they treat initial skills as exogenous and focus

on skill accumulation on the job. Furthermore, their analysis evaluates the labor market transition under the assumption that uncertainty is symmetric across occupations. This paper takes a complementary approach by explicitly focusing on the endogenous pre-labor market training choice and modeling the training-specific uncertainty surrounding AI's ultimate capabilities.

The rest of the paper is organized as follows. Section 2 develops the theoretical model. Section 3 describes the measurement of the AI shock. Section 4 describes the calibration strategy, and Section 5 presents the quantitative simulation results. Section 6 presents robustness checks, using alternative parameter values. Finally, Section 7 concludes.

2 The Model

I consider an economy where agents live for two periods: young ($g = y$) and old ($g = o$). The economy features a finite set of potential training paths. Each training path involves a period of training (when young) followed by a working period (when old). The only exception is the case of no training, which represents a fallback option with no investment in human capital. Training paths may be of two types: a *technical* training path (e.g., plumber, cook) is characterized by a period of apprenticeship as training, while a *cognitive* training path involves a period of studying in college in a certain field before entering the labor market. Let us denote the different training paths by $j \in \{0, 1, \dots, J\}$ ($j = 0$ refers to the case of no training). When young ($g = y$), agents are endowed with a unit of time and choose a training path j . This choice involves an irreversible investment in human capital with uncertain returns. In the working period ($g = o$), agents supply labor inelastically, realize their wage (which depends on the chosen training and labor market conditions), and consume. To better understand the timing of the model, Figure 1 provides a visual representation of the sequence of events. Before AI, workers had to choose their training path without anticipating any AI shocks. After they chose the optimal training, they enter the labor market where they supply their (trained) labor inelastically. In the empirical part, I will use the workers' distribution across training paths in 2022 as the pre-AI steady state. After the introduction of AI, the new generation of workers (ideally students finishing high school after 2022) take into account AI shocks when choosing their optimal training path. After they choose the optimal training, they enter the labor market where they supply their (trained) labor inelastically.

AI Introduction (November 2022)

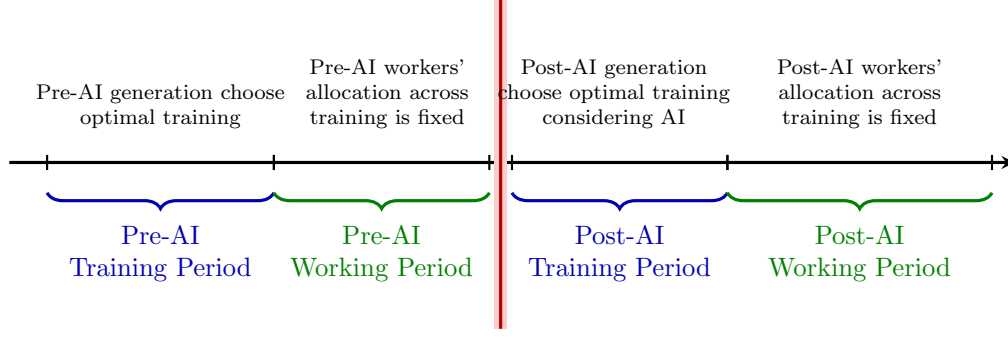


Figure 1: Timeline of the Model.

Notes: The vertical red bar marks the introduction of AI. The pre-AI generation (left) makes training choices and works without anticipating AI shocks. The post-AI generation (right) observes the expected AI shocks ($\mu_{A,j}$, $\mu_{\gamma,j}$) and their uncertainty ($\sigma_{A,j}^2$, $\sigma_{\gamma,j}^2$) before making their irreversible training choice.

2.1 Working Period: Environment

A representative firm demands labor for each training path. The output per worker for training j is

$$Y_j = \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau}^{\frac{1}{\rho}} y_{\tau}^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}$$

where $\mathcal{T}_j$ is the set of relevant tasks for training j , y_{τ} is the output of task τ , and $\theta_{j,\tau}$ is the importance weight of task τ in the production for training j .

The production function for task τ is

$$y_{\tau}(l_{\tau}, k_{\tau}) = \begin{cases} l_{\tau} \gamma_{\tau} & \text{if } \tau \in \mathcal{N}_j \\ l_{\tau} \gamma_{\tau} + \phi_{\tau} k_{\tau}, & \text{if } \tau \in \mathcal{A}_j \end{cases}$$

where $\mathcal{A}_j$ and $\mathcal{N}_j$ are, respectively, the sets of automatable and non-automatable tasks in training j , l_{τ} is the share of time a worker with training j spends in task τ , and k_{τ} is the capital devoted to task τ . Finally, γ_{τ} and ϕ_{τ} are parameters that capture the productivity of labor and capital in task τ . In this framework, automation refers to an expansion of the set $\mathcal{A}_j$, and labor augmentation refers to an increase in γ_{τ} . Thus, in the rest of the paper, I will refer to AI automation when AI can entirely perform tasks that were previously done by humans (i.e., when $\mathcal{A}_j$ expands); on the other hand, I will refer to AI augmentation when AI increases the productivity of human workers without replacing them entirely at the task level (i.e., when γ_{τ} increases).

The problem solved by the representative firm is

$$\begin{aligned}
& \max_{l_{j,\tau}, k_{j,\tau}} L_j \left(p_j Y_j - w_j - R \sum_{\tau \in \mathcal{A}_j} k_{j,\tau} \right) \\
& \text{subject to } Y_j = \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau}^{\frac{1}{\rho}} \tilde{y}_\tau^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \\
& \quad \tilde{y}_\tau = y_\tau(l_{j,\tau}, k_{j,\tau}) \\
& \quad \sum_{\tau \in \mathcal{T}_j} l_{j,\tau} = 1
\end{aligned}$$

where L_j is the share of workers with training j in the labor market. Importantly, L_j is fixed from the previous training period choice and cannot be adjusted until the next period (i.e., the next generation).

Assuming that all automatable tasks are produced with capital, i.e., $\frac{w_j}{\gamma_\tau} > \frac{R}{\phi_\tau}$ for all $\tau \in \mathcal{A}_j$, the labor demand for task τ is zero for $\tau \in \mathcal{A}_j$. Thus, the labor demand is given by

$$l_{j,\tau} = \frac{\theta_{j,\tau} \gamma_\tau^{\rho-1}}{\sum_{\tau' \in \mathcal{N}_j} \theta_{j,\tau'} \gamma_{\tau'}^{\rho-1}}$$

and capital per worker is

$$k_{j,\tau} = \theta_{j,\tau} \phi_\tau^{\rho-1} Y_j \left(\frac{p_j}{R} \right)^\rho$$

As a result, the optimal production per worker in training j is

$$Y_j = \Gamma_j^{\frac{\rho}{1-\rho}} \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \right)^{\frac{1}{\rho-1}}$$

where $\Gamma_j = 1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \left(\frac{R/\phi_\tau}{p_j} \right)^{1-\rho}$ is the labor share for training j .

Finally, since markets are perfectly competitive, the wage is given by

$$w_j = p_j Y_j \Gamma_j = p_j \Gamma_j^{\frac{1}{1-\rho}} \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \right)^{\frac{1}{\rho-1}} \quad (1)$$

2.2 Working Period: Equilibrium

Following Althoff and Reichardt (2026), the price of each occupational good p_j is determined in equilibrium through demand and supply. Labor supply is fixed by the previous period's training choice. The workers, in turn, consume and generate demand for the occupational goods. I assume that demand is characterized by a homothetic

and invertible demand function $D_j(\{p_j\}_{j=0}^J)$ that maps prices p_j into relative demand for each occupational good. Importantly, in the working period, the labor supply is fixed and workers supply their labor inelastically.

Definition (Competitive Equilibrium): Given a distribution of workers across training paths $\{L_j\}_{j=0}^J$ such that $\sum_{j=0}^J L_j = 1$, and the supply of capital $\mathcal{K}$, a competitive equilibrium in this economy is given by a sequence of prices $\{p_0, p_1, \dots, p_J, R\}$, such that:

1. Demand for goods equals supply: $D_j(\{p_j\}_{j=0}^J) = L_j Y_j$
2. Demand for capital equals supply: $\mathcal{K} = \sum_{j=0}^J \sum_{\tau \in \mathcal{A}_j} L_j k_{j,\tau}$
3. Y_j , $l_{j,\tau}$ and $k_{j,\tau}$ are given by the optimal production and capital allocation for each training path j .

2.3 Training Period: Optimal Choice

Before entering the working period, agents must choose their training (or forgo training altogether). Agents are risk-averse, with preferences described by a CRRA utility function $u(w) = \frac{w^{1-\eta}}{1-\eta}$. In what follows, I will use the log wage, so the equivalent utility function over the log wage is $u(\ln w) = \frac{e^{(1-\eta)\ln w}}{1-\eta}$. Individuals are heterogeneous with respect to a vector of net costs $C_i = [C_{i0}, C_{i1}, \dots, C_{iJ}]$, which aggregates all non-pecuniary disutility and direct expenses specific to each training path. For example, if a worker really hates math, the cost of choosing engineering will be huge, whereas if she really loves literature, the cost of choosing humanities will be very low.

A young agent i chooses the training path j that maximizes the discounted expected utility of future earnings, net of the investment cost:

$$V_i^y(C_i) = \max_j \{ \beta \mathbb{E}_t [u(\ln w_j)] - C_{ij} \} \quad (2)$$

where $\beta \in (0, 1)$ is the discount factor and the log-wage for training path j is given by the competitive wage equation 1:

$$\ln w_j = \ln p_j + \frac{1}{1-\rho} \ln \Gamma_j(\mathcal{A}_j) + \frac{1}{\rho-1} \ln \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \right)$$

where $\Gamma_j = 1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \left(\frac{R/\phi_\tau}{p_j} \right)^{1-\rho}$. As in other standard task-based models, automation has an ambiguous impact because it shrinks the set of tasks performed by humans (displacement effect), but it increases productivity in the remaining tasks due to the

cost-savings in the automated tasks (productivity effect); also, it generates a change in the equilibrium prices, which has a direct impact on wages. Similarly, augmentation has an ambiguous impact on wages because it increases productivity of labor, but it changes the equilibrium prices. However, when young, agents do not know what will be the final set of automatable tasks $\mathcal{A}_j$ in each training j , nor which tasks will be augmented. To make this mathematically tractable, I map the set of tasks into a continuum and define a continuous scalar $A_j \in [0, 1]$ representing the share of automated tasks (weighted by importance), and a scalar γ_j representing the average augmentation factor across tasks. Let us suppose their beliefs about the future automation share A_j are well proxied by a normal distribution, $\mathcal{N}(\mu_{A,j}, \sigma_{A,j}^2)$, where $\mu_{A,j}$ is the future expected automation share for training j and $\sigma_{A,j}^2$ captures the uncertainty surrounding future automation in training j . Similarly, for augmentation, workers' beliefs about γ_j are proxied by a normal distribution $\mathcal{N}(\mu_{\gamma,j}, \sigma_{\gamma,j}^2)$, where $\mu_{\gamma,j}$ is the expected augmentation for training j and $\sigma_{\gamma,j}^2$ captures the uncertainty around it. Importantly, all the parameters capturing beliefs are training-specific. Thus, there are some training paths characterized by higher risk of automation and others where augmentation is more likely; also, some training paths could be relatively less subject to both automation and augmentation, but they still can be indirectly impacted through demand spillovers. Moreover, the variance is also training specific and it captures the fact that some training paths are fundamentally more uncertain than others. In the empirical part, I use the existing evidence to calibrate the two moments for each distribution.

Because the log-wage $\ln w_j$ is a non-linear function of both the automated task share A_j and the augmentation factor γ_j , its exact distribution is analytically intractable. To restore tractability, I rely on a first-order Taylor expansion (the Delta method) around the expected shocks. Assuming the automation and augmentation shocks are independent, the linearized log-wage is approximately normally distributed, with expected future log-wage

$$\mathbb{E}[\ln w_j] = \ln p_j(\{L_k, \mu_{A,k}, \mu_{\gamma,k}\}_{k=0}^J) + \frac{1}{1-\rho} \ln \Gamma_j(\mu_{A,j}) + \frac{1}{\rho-1} \ln((1-\mu_{A,j})\mu_{\gamma,j}^{\rho-1})$$

where the endogenous expected price depends on the reallocation of labor supply, and expected automation and augmentation; the expected labor share in each training j depends on expected automation; finally, the last term captures the expected productivity, which depends on expected automation and on the share of tasks that will not be automated. The overall variance is the sum of the variances of the underlying shocks,

weighted by the squared sensitivities:

$$Var(\ln w_j) = \left(\frac{\partial \ln w_j}{\partial A_j} \Big|_{\mu_{A,j}, \mu_{\gamma,j}} \right)^2 \sigma_{A,j}^2 + \left(\frac{\partial \ln w_j}{\partial \gamma_j} \Big|_{\mu_{A,j}, \mu_{\gamma,j}} \right)^2 \sigma_{\gamma,j}^2$$

Given this normal approximation for the log-wage, the expected utility can be evaluated exactly using the moment-generating function of the normal distribution:

$$\mathbb{E}[u(\ln w_j)] = \frac{1}{1-\eta} \exp \left((1-\eta)\mathbb{E}(\ln w_j) + \frac{(1-\eta)^2}{2} Var(\ln w_j) \right)$$

The certainty equivalent Ω_j is defined as the guaranteed wage that yields the same expected utility, $u(\Omega_j) = \mathbb{E}[u(\ln w_j)]$. Equating the two yields:

$$\frac{e^{(1-\eta)\Omega_j}}{1-\eta} = \frac{1}{1-\eta} \exp \left((1-\eta)\mathbb{E}(\ln w_j) + \frac{(1-\eta)^2}{2} Var(\ln w_j) \right)$$

Canceling the constant term, taking logs, and rearranging provides the exact certainty equivalent for training path j :

$$\Omega_j = \underbrace{\ln p_j + \frac{1}{1-\rho} \ln \Gamma_j + \frac{1}{\rho-1} \ln ((1-\mu_{A,j})\mu_{\gamma,j}^{\rho-1})}_{\mathbb{E}(\ln w_j)} - \underbrace{\frac{\eta-1}{2} Var(\ln w_j)}_{\text{Risk premium}}$$

which formalizes exactly how agents penalize uncertainty depending on their degree of risk aversion η . Importantly, this model creates a training-specific risk premium that is linked with the relative uncertainty regarding future automation and augmentation due to AI in each training path.

Substituting $\mathbb{E}[u(\ln w_j)]$ with $u(\Omega_j)$, the agent's objective function becomes:

$$V_i^y(C_i) = \max_j \{ \beta u(\Omega_j) - C_{ij} \}$$

To render the discrete choice problem analytically tractable, I apply a first-order Taylor expansion to the utility of the certainty equivalent $u(\Omega_j)$ around a baseline average log-wage $\bar{\mu}$. Substituting $u(\Omega_j) \approx u(\bar{\mu}) + u'(\bar{\mu})(\Omega_j - \bar{\mu})$ into the objective, the maximization problem simplifies to:

$$\max_j \{ \beta \Omega_j - C_{ij}^* \} \tag{3}$$

where $C_{ij}^* \equiv C_{ij}/u'(\bar{\mu})$ is the normalized cost parameter. Notice that, although we are linearizing the utility function, the risk aversion coefficient η still enters into the decision rule through the risk premium in the certainty equivalent Ω_j .

I assume that the normalized cost is determined by a training-specific common average cost, $\bar{C}_j^*$, and an idiosyncratic preference component, such that $C_{ij}^* = \bar{C}_j^* - \epsilon_{ij}$. Assuming the idiosyncratic shocks ϵ_{ij} are independently and identically distributed following a Type I Extreme Value (Gumbel) distribution with location parameter zero and scale parameter ζ , the aggregate market share of labor choosing training path j is mapped by the standard multinomial conditional logit formulation (McFadden (1972)):

$$L_j^* = \frac{\exp((\beta\Omega_j - \bar{C}_j^*)/\zeta)}{\sum_{h=0}^J \exp((\beta\Omega_h - \bar{C}_h^*)/\zeta)} \quad (4)$$

2.4 Theoretical Results and Equilibrium Properties

Using the structural discrete choice framework derived above, this section establishes the comparative statics of the model with respect to the four parameters of the agents' beliefs about the automation and augmentation shocks: the expected levels ($\mu_{A,j}$ and $\mu_{\gamma,j}$) and their respective uncertainties ($\sigma_{A,j}^2$ and $\sigma_{\gamma,j}^2$).

Proposition 1 (Wage Response to Labor Supply). *Because output per worker Y_j is independent of L_j , an increase in the labor supply L_j for training path j (i.e., more workers choosing training j) raises the aggregate supply of final good produced with training j . Given the homothetic demand function D_j , this generates a fall in the equilibrium price p_j , which translates one-for-one into a fall in the expected log-wage:*

$$\frac{\partial \mathbb{E}[\ln w_j]}{\partial L_j} = \frac{\partial \ln p_j}{\partial L_j} < 0 \quad (5)$$

Conversely, a decrease in L_j raises the equilibrium price p_j and therefore raises $\mathbb{E}[\ln w_j]$. Expected wages are thus strictly decreasing in labor supply.

Intuition: When more workers crowd into a training path, they collectively produce more of the associated good. The resulting excess supply depresses its price, and since wages equal the value of marginal product, expected wages fall. The reverse holds when workers leave a path: scarcity drives up the price and lifts expected wages. This mechanism is the central general-equilibrium channel through which all belief shocks (expected automation, expected augmentation, and their uncertainties) translate into wage adjustments.

The result in Proposition 1 allows us to map how exogenous changes in beliefs affect actual equilibrium wages. Note that the derivatives derived from the conditional logit formula in Proposition 2 represent the *direct* (or partial equilibrium) shifts in labor supply holding expected wages and prices constant. In general equilibrium, these initial

shifts trigger price adjustments that ultimately dampen the final reallocation; however, this effect can never reverse the direction of the response.

Proposition 2 (Flight from Uncertainty). *An increase in automation uncertainty ($\uparrow \sigma_{A,j}^2$) or augmentation uncertainty ($\uparrow \sigma_{\gamma,j}^2$) for training path j leads to a decrease in the equilibrium employment L_j^* :*

$$\frac{\partial L_j^*}{\partial \sigma_{A,j}^2} = -\frac{\beta(\eta-1)}{2\zeta} \left(\frac{\partial \ln w_j}{\partial A_j} \right)^2 L_j^*(1-L_j^*) < 0 \quad (6)$$

$$\frac{\partial L_j^*}{\partial \sigma_{\gamma,j}^2} = -\frac{\beta(\eta-1)}{2\zeta} \left(\frac{\partial \ln w_j}{\partial \gamma_j} \right)^2 L_j^*(1-L_j^*) < 0 \quad (7)$$

Intuition: As a career path becomes more unpredictable due to exposure to AI, risk-averse individuals systematically flee to safer alternatives. The magnitude of this structural flight depends directly on their degree of risk aversion (η) and on the scale parameter ζ .

Proposition 3 (Endogenous Risk Premium). *An increase in either automation or augmentation uncertainty for training path j ($\uparrow \sigma_{A,j}^2, \uparrow \sigma_{\gamma,j}^2$) raises the expected wages $\mathbb{E}(\ln w_j)$. This reflects an endogenous risk premium driven by labor scarcity:*

$$\frac{\partial \mathbb{E}(\ln w_j)}{\partial \sigma_{A,j}^2} = \underbrace{\frac{\partial \mathbb{E}(\ln w_j)}{\partial p_j}}_{(>0)} \underbrace{\frac{\partial p_j}{\partial L_j}}_{(<0)} \underbrace{\frac{\partial L_j^*}{\partial \sigma_{A,j}^2}}_{(<0)} > 0, \quad \frac{\partial \mathbb{E}(\ln w_j)}{\partial \sigma_{\gamma,j}^2} = \underbrace{\frac{\partial \mathbb{E}(\ln w_j)}{\partial p_j}}_{(>0)} \underbrace{\frac{\partial p_j}{\partial L_j}}_{(<0)} \underbrace{\frac{\partial L_j^*}{\partial \sigma_{\gamma,j}^2}}_{(<0)} > 0 \quad (8)$$

Intuition: This characterizes a novel theoretical mechanism of the model. Uncertainty reduces the share of labor choosing training path j (see Proposition 2). The resulting labor scarcity pushes up the equilibrium price of associated good, mechanically increasing the realized wage for those who remain. This can be seen as a general equilibrium risk premium that is endogenously generated by the labor market reallocation, rather than the standard risk premium arising from investment in a risky asset.

3 Measuring the AI Shocks

3.1 Measuring the AI Shocks: Methodology

To implement the model quantitatively, we need to measure the expected values ($\mu_{A,j}$ and $\mu_{\gamma,j}$) and the uncertainties ($\sigma_{A,j}^2$ and $\sigma_{\gamma,j}^2$) of the automation and augmentation shocks for each training path j . I compute these moments by categorizing the tasks comprising each training path based on their current and potential AI exposure.

This categorization relies on two main sources: the actual task penetration of AI (Massenkoff and McCrory, 2026), and the theoretical automatability and augmentation potential of tasks (Althoff and Reichardt, 2026).

Massenkoff and McCrory (2026) construct a measure of current task penetration of AI, by using recent Anthropic data. The task penetration is zero if, at the moment, AI is not used in that task. If instead AI is already used in that task, its penetration is positive, with values closer to 1 meaning that AI is more automating than augmenting that task (see Massenkoff and McCrory (2026) for more details).

Althoff and Reichardt (2026) provides two measures: theoretical automatability and theoretical augmentation. Theoretical automatability is either 1 if the task is potentially automatable using generative AI, or 0 otherwise. Theoretical augmentation is 1 if the task is not augmentable with AI and greater than 1 if it is potentially augmentable; for example a value of 1.1 means that AI can boost labor productivity by 10% in that task (see Althoff and Reichardt (2026) for more details).

First, I define the *current status* of a task:

- **Automated:** The actual task penetration is greater than or equal to 0.9 and the task is automatable (automatable = 1).
- **Augmented:** The actual task penetration is greater than 0, the task is not automatable (automatable = 0). Alternatively, the task is automatable (automatable = 1), but the current task penetration is less than 0.9.
- **Neither:** If the actual task penetration is zero, the task is currently neither automated nor augmented.

Second, I define the *potential status* of a task:

- If a task is already automated, its potential status is automated (with no uncertainty).
- If a task is already augmented and it is not automatable, its potential status is augmented (with no uncertainty).
- **Potentially automatable:** The task’s actual penetration is lower than 0.9, but the task is theoretically automatable (automatable = 1).
- **Potentially augmentable:** The task’s actual penetration is zero, it is not automatable (automatable = 0), and its augmentation factor is greater than 1.
- **Not affected by AI:** The task’s actual penetration is zero, it is not automatable (automatable = 0), and its augmentation factor is exactly 1.

To formalize this aggregation, let $\mathcal{T}_j$ be the set of tasks associated with training path j , and let $\theta_{j,\tau}$ denote the importance weight of task τ (such that $\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} = 1$). Let $a_\tau \in \{0, 1\}$ be a random variable indicating whether task τ will eventually be automated, and let $b_\tau \in \{0, 1\}$ be a random variable indicating whether it will be augmented.

For automation, if a task is already automated, its future outcome is certain: $\mathbb{E}[a_\tau] = 1$ and $\text{Var}(a_\tau) = 0$. Conversely, if it is not even potentially automatable, $\mathbb{E}[a_\tau] = 0$ and $\text{Var}(a_\tau) = 0$. The uncertainty lies entirely with the tasks that are *potentially automatable* but not yet automated. For these tasks in the “gap”, I assume a probability $p_A \in (0, 1)$ that the theoretical automatability translates into actual automation. Thus, for tasks in this gap, $\mathbb{E}[a_\tau] = p_A$ and $\text{Var}(a_\tau) = p_A(1 - p_A)$.

The expected automation share $\mu_{A,j}$ for training path j is the weighted sum of these task-level expectations:

$$\mu_{A,j} = \sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} \mathbb{E}[a_\tau] = S_{A,j}^{cur} + p_A S_{A,j}^{pot} \quad (9)$$

where $S_{A,j}^{cur}$ is the currently automated (weighted) share and $S_{A,j}^{pot}$ is the potentially automatable (weighted) share (excluding the currently automated tasks). To formalize the possibility that AI success is correlated across tasks within the same training path, I introduce a correlation parameter $\kappa \in [0, 1]$. A value of $\kappa = 0$ implies independent task automation, whereas $\kappa = 1$ implies perfect correlation (AI automates either all potential tasks or none). The variance of the automation shock then decomposes into a convex combination of the independent and perfectly correlated variances:

$$\sigma_{A,j}^2 = p_A(1 - p_A) \left[(1 - \kappa) \sum_{\tau \in S_{A,j}^{pot}} \theta_{j,\tau}^2 + \kappa \left(\sum_{\tau \in S_{A,j}^{pot}} \theta_{j,\tau} \right)^2 \right] \quad (10)$$

where $S_{A,j}^{pot}$ is the set of tasks that are potentially automatable but not yet automated. First, notice that the total variance is strictly increasing in the total automation gap $\sum_{\tau \in S_{A,j}^{pot}} \theta_{j,\tau}$; therefore, training paths with a higher unrealized potential for automation inherently face higher uncertainty. Second, the term $\sum_{\tau \in S_{A,j}^{pot}} \theta_{j,\tau}^2$ acts as a Herfindahl-Hirschman Index (HHI) of task concentration. This implies that, especially when $\kappa < 1$, holding the gap share constant, training paths that are highly specialized in a few uncertain tasks face significantly more risk than paths with a diversified portfolio of many small tasks.

A symmetric logic applies to augmentation. Let γ_τ be the theoretical augmentation factor for task τ (where $\gamma_\tau > 1$ means the task is augmentable, and $\gamma_\tau = 1$ means no

gain). For tasks that are strictly augmented (already augmented and *not* potentially automatable), their outcome is certain: $\mathbb{E}[b_\tau] = 1$ and $\text{Var}(b_\tau) = 0$. Conversely, if the task is not even potentially augmentable, $\mathbb{E}[b_\tau] = 0$ and $\text{Var}(b_\tau) = 0$.

The uncertainty in augmentation, captured by the set $S_{\gamma,j}^{pot}$, arises from two distinct sources. First, tasks that are *potentially augmentable* but not yet augmented have a probability p_γ of realizing this productivity boost. Second, tasks that are *already augmented* but also *potentially automatable* face the risk of full automation; if automation succeeds (with probability p_A), the human is replaced and the augmentation factor effectively drops back to 1. Thus, they only have a probability $1 - p_A$ of remaining augmented. In general p_γ and $1 - p_A$ need not coincide, so the two sources of uncertainty carry different Bernoulli parameters. Let $S_{\gamma,j}^{new}$ denote the set of tasks that are potentially augmentable but not yet augmented, and $S_{\gamma,j}^{surv}$ the set of tasks that are already augmented but also potentially automatable. For a task $\tau \in S_{\gamma,j}^{new}$, $\mathbb{E}[b_\tau] = p_\gamma$ and $\text{Var}(b_\tau) = p_\gamma(1 - p_\gamma)$; for a task $\tau \in S_{\gamma,j}^{surv}$, $\mathbb{E}[b_\tau] = 1 - p_A$ and $\text{Var}(b_\tau) = p_A(1 - p_A)$. The expected augmentation factor for training path j is:

$$\mu_{\gamma,j} = \sum_{\tau \in T_j} \theta_{j,\tau} (1 + \mathbb{E}[b_\tau](g_\tau - 1)) \quad (11)$$

To derive the variance, let $\nu_\tau \equiv \text{Var}(b_\tau)$ denote the task-specific Bernoulli variance (which equals $p_\gamma(1 - p_\gamma)$ for tasks in $S_{\gamma,j}^{new}$ and $p_A(1 - p_A)$ for tasks in $S_{\gamma,j}^{surv}$). The variance of the augmentation factor, allowing for cross-task correlation κ , is:

$$\sigma_{\gamma,j}^2 = (1 - \kappa) \sum_{\tau \in S_{\gamma,j}^{pot}} \theta_{j,\tau}^2 (g_\tau - 1)^2 \nu_\tau + \kappa \sum_{\tau, \tau' \in S_{\gamma,j}^{pot}} \theta_{j,\tau} \theta_{j,\tau'} (g_\tau - 1)(g_{\tau'} - 1) \sqrt{\nu_\tau \nu_{\tau'}} \quad (12)$$

where $S_{\gamma,j}^{pot} = S_{\gamma,j}^{new} \cup S_{\gamma,j}^{surv}$. Notice that when $p_\gamma = 1 - p_A$, all tasks share a common $\nu_\tau = p_\gamma(1 - p_\gamma)$, and this expression collapses to the familiar single-probability form.

Just as with automation, this variance is driven by two key factors: the size of the potential augmentation gap (meaning paths with a higher potential productivity boost naturally face higher uncertainty) and the concentration of that potential across a few specific tasks.

3.2 Measuring the AI Shocks: Results

For the baseline empirical implementation, I set the probability of automation $p_A = 0.5$ and the probability of augmentation $p_\gamma = 0.5$. This means that if automation or augmentation is theoretically possible, agents believe there is a 50% chance that it will actually occur. I also run an alternative specification where these probabilities are higher:

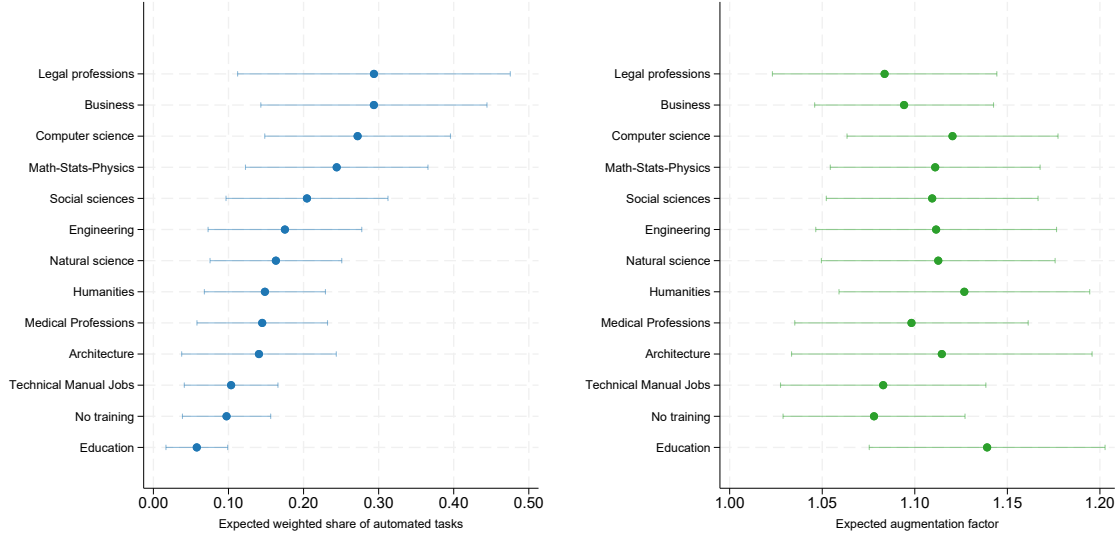


Figure 2: Expected Automation and Augmentation across Training Paths (with 1 standard deviation confidence interval).

Notes: This figure compares the expected weighted share of automated tasks ($\mu_{A,j}$, left panel) and expected augmentation factor ($\mu_{\gamma,j}$, right panel, 1 means no augmentation, 1.1 means a 10% productivity boost) for each training path j , computed by aggregating task-level AI exposure using importance weights. The dots represent expected values and the error bars span ± 1 standard deviation ($\sigma_{A,j}$ and $\sigma_{\gamma,j}$), capturing the uncertainty surrounding each shock. Expected values and variances are constructed by combining current AI task penetration (Massenkoff and McCrory (2026)) with theoretical automatability and augmentation potential (?). Uncertainty arises exclusively from tasks in the “gap”—those that are theoretically automatable (or augmentable) but not yet automated (or augmented)—each assigned a probability $p_A = 0.5$ ($p_\gamma = 0.5$) of eventual realization, with cross-task correlation $\kappa = 0.5$.

$p_A = 0.8$ and $p_\gamma = 0.8$. The results are qualitatively similar; the only difference is that workers react more strongly to the AI shocks when they believe that automation and augmentation are more likely to occur. The correlation parameter is set to $\kappa = 0.5$ in the baseline, the exact midpoint between the two unrealistic extremes of independence ($\kappa = 0$) and perfect correlation ($\kappa = 1$). Figure 2 displays the expected automation ($\mu_{A,j}$) and expected augmentation ($\mu_{\gamma,j}$) shocks across different training paths in the baseline. Furthermore, the confidence intervals capture differences in uncertainty across training paths. For instance, the training paths with the most automation uncertainty are legal professions and business, because many of their tasks are potentially automatable, but not yet automated. Computer science has a lower uncertainty compared to them (still relatively high) because some of its tasks have already been automated. On the other hand, paths like technical manual jobs and education have a very low expected automation (and uncertainty) because their tasks are more difficult to automate. As far as augmentation is concerned, both expectations and uncertainty are similar across training paths; as a result, its role in shaping labor reallocation is more limited compared to automation. In general, the training paths most exposed to automation are those that are more abstract and cognitive (legal, business, computer science), while the paths least exposed to automation are those that require manual, physical skills (technical manual, no training) or social interactions (education, medical). Lastly, Figure 3 shows the same figures across training paths in the alternative scenario where $p_A = 0.8$ and $p_\gamma = 0.8$. The results are qualitatively similar to the baseline, but the expected shocks are higher, reflecting the agents' belief that AI is more likely to be eventually successful.

4 Calibration

The quantitative implementation of the model proceeds in two main stages. First, I calibrate the model to a pre-AI baseline equilibrium (representing the year 2022) where AI shocks are not yet anticipated. This allows me to recover the fundamental structural parameters of the economy, particularly the unobserved training costs. Second, I introduce the measured AI shocks (both expected productivity and uncertainty) derived in Section 3 to simulate the new post-AI equilibrium and decompose its effects on employment and wages. The main assumption is that AI does not affect the fundamental structural parameters of the economy (e.g., training disutility costs, preferences, and elasticities). Obviously, this is a strong assumption, but if AI does not change drastically the structural parameters, at least in relative terms, the implications of the model are robust to this assumption.

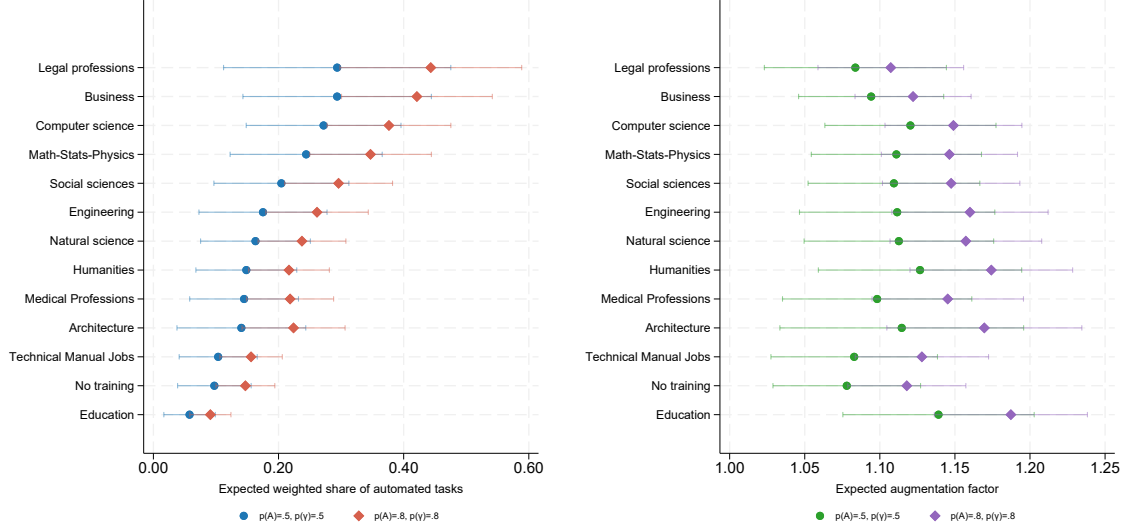


Figure 3: Expected Automation and Augmentation across Training Paths (with 1 standard deviation confidence interval): Baseline vs Alternative Scenario.

Notes: This figure compares the expected weighted share of automated tasks ($\mu_{A,j}$, left panel) and expected augmentation factor ($\mu_{\gamma,j}$, right panel, 1 means no augmentation, 1.1 means a 10% productivity boost) for each training path j in two scenarios, computed by aggregating task-level AI exposure using importance weights. The dots represent expected values and the error bars span ± 1 standard deviation ($\sigma_{A,j}$ and $\sigma_{\gamma,j}$), capturing the uncertainty surrounding each shock. Expected values and variances are constructed by combining current AI task penetration (Massenkoff and McCrory (2026)) with theoretical automatability and augmentation potential (?). Uncertainty arises exclusively from tasks in the “gap”—those that are theoretically automatable (or augmentable) but not yet automated (or augmented). In the baseline scenario (blue and green dots), each task in the gap is assigned a probability $p_A = 0.5$ ($p_\gamma = 0.5$) of eventual realization, whereas in the alternative scenario (red and purple diamonds), each task in the gap is assigned a probability $p_A = 0.8$ ($p_\gamma = 0.8$) of eventual realization. The cross-task correlation is set to $\kappa = 0.5$ in both scenarios.

4.1 Pre-AI Baseline Equilibrium (2022)

In the pre-AI steady state, agents do not anticipate the arrival of Artificial Intelligence. Consequently, their expectations about future automation and augmentation are zero ($\mu_{A,j} = \mu_{\gamma,j} = 0$), and there is no uncertainty regarding these shocks ($\sigma_{A,j}^2 = \sigma_{\gamma,j}^2 = 0$). Without uncertainty, the risk premium is zero, and the certainty equivalent Ω_j collapses exactly to the expected current log-wage: $\Omega_j^{pre} = \ln w_j^{2022}$.

Using the optimal labor supply derived in Equation (4), the employment share for training path j in the pre-AI equilibrium is given by:

$$L_j^{2022} = \frac{\exp((\beta \ln w_j^{2022} - \bar{C}_j^*)/\zeta)}{\sum_{h=0}^J \exp((\beta \ln w_h^{2022} - \bar{C}_h^*)/\zeta)} \quad (13)$$

where L_j^{2022} and w_j^{2022} are the observed employment shares and median wages from the 2022 American Community Survey (ACS) and Bureau of Labor Statistics (BLS) data, across the 13 training paths.

By observing the actual labor allocations and wages, I can invert this mapping to exactly recover the unobserved normalized training costs $\bar{C}_j^*$. Normalizing the cost of the no-training option ($j = 0$) to zero ($\bar{C}_0^* = 0$), the structural cost parameters are identified as:

$$\bar{C}_j^* = \beta(\ln w_j^{2022} - \ln w_0^{2022}) - \zeta(\ln L_j^{2022} - \ln L_0^{2022}) \quad (14)$$

On the labor demand side, following the procedure outlined in Althoff and Reichardt (2026), I assume that occupational goods are substituted with a constant elasticity of substitution (CES) ϵ , so that demand for occupation j satisfies $D_j(\mathbf{p}) \propto \alpha_j p_j^{-\epsilon}$. I set the elasticity of substitution between occupations $\epsilon = 1.57$ —the midpoint between 1.81 (Burstein et al., 2019) and 1.34 (Caunedo et al., 2023). The CES importance weights α_j are then identified from observed occupational wage shares and the estimated occupational prices because for any two occupations i and j :

$$\frac{\alpha_i}{\alpha_j} = \left(\frac{p_i}{p_j}\right)^{\epsilon-1} \times \frac{\text{Wage share of occupation } i}{\text{Wage share of occupation } j} \quad (15)$$

Because no tasks are automated in the pre-AI baseline economy, the labor share is exactly one ($\Gamma_j = 1$) and output per worker is normalized to one ($Y_j = 1$). Consequently, the competitive wage equation $w_j = p_j Y_j \Gamma_j$ simplifies to $p_j = w_j$. Thus, I can directly substitute the observed 2022 wages w_j^{2022} for the prices p_j in the equation above to perfectly calibrate the demand shifters α_j .

4.2 Externally Calibrated Parameters

To simulate the model quantitatively, I calibrate several fundamental parameters directly from the recent literature on task-based models, relying primarily on the estimates provided by Althoff and Reichardt (2026).

Macro Elasticities and Preferences: I set the elasticity of substitution across tasks within an occupation to $\rho = 0.49$, implying that tasks are complementary. For the agents' preferences, the annual discount factor is set to $\beta = 0.813$, corresponding to an annual discount rate of 3% and considering 7 years (from 18 to 25) for the training period, and the coefficient of relative risk aversion is set to a baseline value of $\eta = 5$. In the results section, I additionally test the sensitivity of the model to a conservative scenario ($\eta = 2$) and a high risk aversion scenario ($\eta = 10$). Finally, ζ captures the rigidity of the training choice: a lower ζ means that agents are more sensitive to differences in expected wages and training costs, while a higher ζ implies that agents are more indifferent to these differences. In the baseline scenario, I use $\zeta = 0.053$, following Althoff and Reichardt (2026) who calibrated the Gumbel dispersion parameter for idiosyncratic occupational preferences. However, since here the focus is on training rather than occupation, I will provide also the results using a higher value of $\zeta = 0.3$.

Automation and Augmentation: The expected shocks (μ) and uncertainties (σ^2) defined in Section 3 require calibrating the probability that a task in the “gap” is eventually exposed to AI. To reflect a state of maximum agnostic uncertainty regarding the future of these tasks, in the baseline scenario, I set the adoption probabilities to $p_A = 0.5$ and $p_\gamma = 0.5$. Furthermore, I use the task-specific augmentation factors g_τ from (Althoff and Reichardt, 2026). I also provide results for an alternative scenario where the adoption probabilities are higher ($p_A = 0.8$ and $p_\gamma = 0.8$).

Capital-Labor Substitution: While the pre-AI baseline economy assumes no tasks are automated by generative AI (so the initial labor share is exactly $\Gamma_j^{pre} = 1$), simulating the post-AI economy requires quantifying how the labor share responds to the automation shock. Rather than calibrating task-specific capital rental rates and productivities, I follow Althoff and Reichardt (2026) and assume that the cost savings of using AI instead of labor for automatable tasks is 27%, yielding a relative cost of capital $\chi = 0.73$. Using their structural derivation, the post-AI labor share for any training path j can be approximated as:

$$\Gamma_j^{post} \approx \frac{\chi^{\rho-1} (1 - \mu_{A,j})}{\mu_{A,j} + \chi^{\rho-1} (1 - \mu_{A,j})} \quad (16)$$

where $\mu_{A,j}$ is the expected total share of tasks automated by AI in training path j (as

derived in Section 3).

4.3 Simulating the Post-AI Equilibrium

Once the structural parameters are pinned down, I introduce the AI shocks. For each training path j , I plug in the expected values $(\mu_{A,j}, \mu_{\gamma,j})$ and variances $(\sigma_{A,j}^2, \sigma_{\gamma,j}^2)$ computed in Section 3.

These shocks alter the certainty equivalent Ω_j^{post} for each training path through both the direct expected productivity shifts and the endogenous risk premiums. The new post-AI equilibrium is defined as the fixed point where:

1. Risk-averse agents optimally reallocate across training paths based on the new certainty equivalents, forming the post-AI labor supply L_j^{post} .
2. The new labor supply alters the production of occupational goods, which in turn adjusts the equilibrium prices p_j^{post} according to the CES demand function.
3. Expected log-wages $\mathbb{E}[\ln w_j]^{post}$ adjust endogenously to clear the market.

By comparing the post-AI equilibrium to the pre-AI baseline, we can quantify the total effect of AI on employment and expected wages, and further decompose these effects to understand the contributions of the different channels of the model.

5 Results and Decomposition

This section presents the quantitative predictions of the calibrated model, in the baseline scenario. To isolate the contribution of each economic mechanism, the full post-AI equilibrium is decomposed sequentially into four channels: the *expected automation* effect, the *expected augmentation* effect, and the *uncertainty* effects associated with both automation and augmentation shocks.

5.1 Employment Reallocation

Figure 4 presents the decomposition of AI-induced employment shifts across the 13 training paths.

The *expected automation* channel drives the primary structural transformation. The training paths most exposed to automation—such as business, computer science, and legal professions—are expected to be chosen less by the new generation of workers, with legal professions losing -12% of their workforce only because of the automation threat. Conversely, paths with high physical or interpersonal requirements—such as technical

manual jobs and education, as well as the “no training” residual category, experience positive labor inflows. The *expected augmentation* channel plays a comparatively minor role in employment reallocation, as human-enhancing technologies boost output across most sectors relatively symmetrically without dramatically shifting relative labor demand.

Crucially, the *uncertainty* over automation triggers a secondary reallocation. Risk-averse agents actively flee the highly uncertain training paths (e.g., business and legal professions). This flight from uncertainty amplifies the initial expected automation shock, accelerating the employment drop in exposed careers and crowding workers into the perceived safer options. Uncertainty regarding augmentation has virtually zero impact on employment, again because the augmentation uncertainty is relatively similar across training paths.

Overall, the model predicts a substantial reallocation of labor across training paths, with the most exposed paths (legal professions, business, and computer science) losing up to -16% of their workforce, while the safer paths (technical manual jobs, education, and no training) gain up to $+11\%$.

5.2 Wage Effects

The wage decomposition (Figure 5) reveals the power of general equilibrium forces. The expected AI shocks—both automation and augmentation—are unequivocally beneficial for aggregate productivity, generating an across-the-board increase in expected wages ranging from $+4.2\%$ to $+5.4\%$ from automation, and $+9.5\%$ from augmentation. The primary driver of this result is the reallocation of labor across training paths. If we take labor supply as fixed to the pre-AI baseline, the expected wage impact driven by automation is dramatically diverse. Training paths very exposed to automation (legal, business, and computer science) would see their expected wages drop by more than 2% , whereas the safer paths (technical manual jobs, education, and no training) would gain up to $+11\%$ only because of expected automation. This is intuitive: automation boosts productivity in the non-automated tasks; therefore if workers don’t reallocate, training paths that are less exposed to automation would gain a lot (low displacement effect), whereas the more exposed paths would lose because of the displacement effect. The fact that individuals reallocate toward positively impacted paths, however, erodes the expected wage gains in those paths and generates a scarcity effect in the exposed paths. These results highlight the importance of accounting for the labor supply responses when evaluating the impact of AI on wages.

The impact of uncertainty is qualitatively similar. Risk-averse agents flee the highly uncertain training paths, which reduces employment in those paths. The labor scarcity

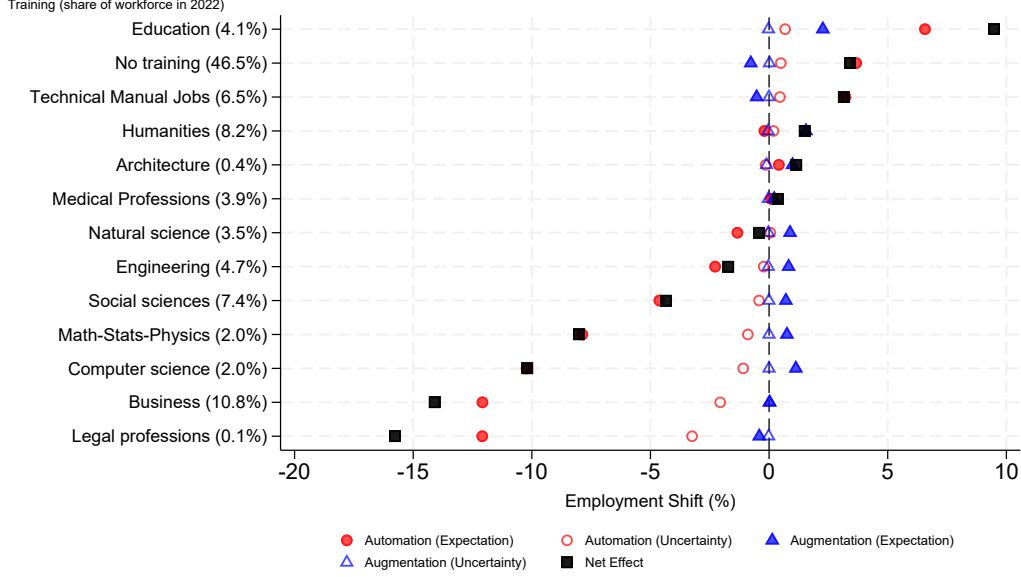


Figure 4: Decomposition of AI-induced employment shifts across training paths.

Notes: This figure decomposes, in the baseline scenario, the percentage change in employment shares (relative to the pre-AI 2022 baseline) into four channels: expected automation (red solid circles), uncertainty over automation (red hollow circles), expected augmentation (blue solid triangles), and uncertainty over augmentation (blue hollow triangles). The total net effect is shown as black squares. Expected automation is the primary driver of labor reallocation, shifting workers away from highly exposed cognitive training paths—legal professions (−16%), business, and computer science—toward paths with high physical or interpersonal requirements, such as technical manual jobs and education, as well as the no-training option. Automation uncertainty amplifies this reallocation: risk-averse agents actively flee the most uncertain paths (e.g., business and legal professions), further crowding workers into perceived safer options. Expected augmentation and its uncertainty play a comparatively minor role in employment reallocation, because augmentation boosts output relatively symmetrically across training paths. See Section 4 for details on the baseline calibration. See Section 3 for details on the expected AI shocks and their uncertainty.

generated by this flight to safety, however, creates an endogenous risk premium that drives up expected wages in the exposed paths. Quantitatively, this channel is less pronounced, but it is still economically significant. The most uncertain training paths (legal and business) see their expected wages increase by +2.1% and +1.3%, respectively, only because of the uncertainty-driven labor scarcity.

The impact of augmentation is strongly positive and relatively uniform across training paths—an increase of roughly +9.5%. This is because augmentation boosts productivity in all training paths, and the expected augmentation shock is relatively similar across paths. The uncertainty over augmentation has a negligible effect on wages, as the uncertainty is also relatively similar across training paths.

The final (optimistic) result is a remarkably uniform wage increase of roughly +14% across all training paths. The only exceptions are legal professions and business, which

see a slightly higher increase in their wages due to the endogenous risk premium.

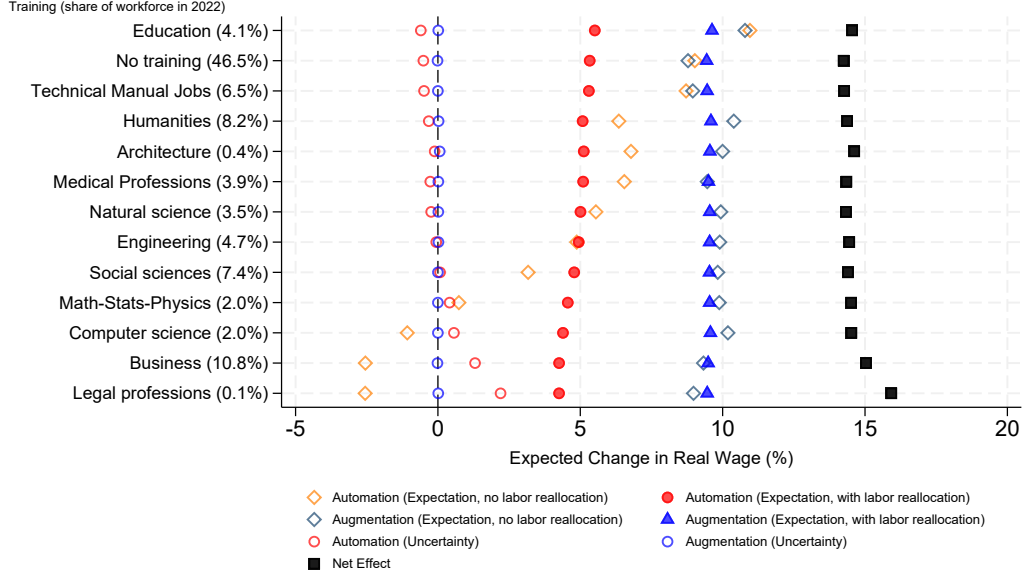


Figure 5: Decomposition of AI-induced expected wage changes across training paths.

Notes: This figure decomposes, in the baseline scenario, the percentage change in expected log-wages into six components: expected automation with no labor reallocation (orange hollow diamond), expected automation with labor reallocation (red solid circle), uncertainty over automation (red hollow circles), expected augmentation with no labor reallocation (navy hollow diamond), expected augmentation with labor reallocation (blue solid circle), and uncertainty over augmentation (blue hollow triangles). The total net wage change is shown as black squares. In contrast to the employment effects, expected AI shocks are unequivocally beneficial for wages: expected automation raises expected wages by +4.2% to +5.4%, and expected augmentation by approximately +9.5%, yielding a total increase of roughly +14% across all training paths. This uniform wage increase is driven by a general equilibrium scarcity channel: as workers reallocate away from highly exposed paths, the resulting labor scarcity pushes up equilibrium prices and wages in those paths, while congestion compresses wages in safer paths; the two forces nearly cancel, producing a remarkably uniform wage gain. Automation uncertainty generates an additional endogenous risk premium of up to +2.3% (legal professions: +2.1%; business: +1.3%), as the flight from uncertain paths creates further labor scarcity that raises expected wages beyond what expected productivity effects alone would predict. Augmentation uncertainty has a negligible effect on wages. See Section 4 for details on the baseline calibration. See Section 3 for details on the expected AI shocks and their uncertainty.

To further highlight the endogenous risk premium, Figure 6 plots the expected wage changes against the variance of the AI shock across the different training paths. The evident positive relationship confirms that paths facing higher technological uncertainty compensate workers with higher expected wages. This highlights the general equilibrium effect: as risk-averse workers flee highly volatile training paths, the resulting labor scarcity mechanically drives up the expected wages in those training paths.

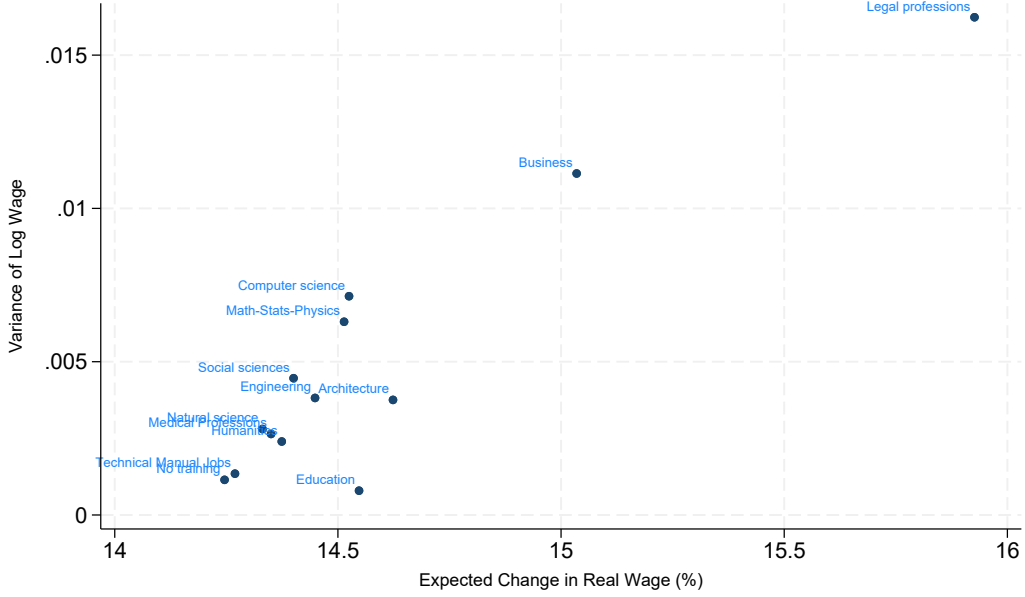


Figure 6: Expected Wage Changes vs. AI Shock Variance across Training Paths.

Notes: Each point represents a training path. The vertical axis measures the total variance of the log wage ($Var(\ln w_j)$), which combines the variance of the automation and augmentation shocks weighted by their respective wage sensitivities (see Section 2.3). The horizontal axis measures the total percentage change in expected log-wages relative to the pre-AI 2022 baseline. The positive relationship confirms the model’s prediction of an endogenous risk premium: training paths facing higher AI-related uncertainty (e.g., legal professions and business) experience larger expected wage increases. This premium is a pure general equilibrium effect driven by labor scarcity—risk-averse agents flee highly uncertain paths (Proposition 2), reducing labor supply and mechanically pushing up equilibrium prices and wages for those who remain (Proposition 1)—rather than a standard compensation for holding a risky asset. See Section 4 for details on the baseline calibration. See Section 3 for details on the expected AI shocks and their uncertainty.

6 Robustness

In this section, I test the robustness of the main results to alternative parameterizations of the model. I focus on three key parameters: the risk aversion coefficient η , the Gumbel dispersion parameter ζ , and the probabilities of automation and augmentation (p_A and p_γ). The results are qualitatively similar across all alternative specifications, confirming the robustness of the main findings.

As far as the risk aversion coefficient η is concerned, I test a conservative scenario with $\eta = 2$ and a high risk aversion scenario with $\eta = 10$. Figure A1 shows that labor reallocation is largely similar across the three scenarios, with the most exposed paths (legal professions, business, and computer science) losing up to -20% of their workforce, while the safer paths (technical manual jobs, education, and no training) gain up to $+10\%$. The main difference is that higher risk aversion amplifies the flight from uncertainty, leading to slightly larger employment shifts in the most exposed paths. As

a result, the endogenous risk premium is also larger in the high risk aversion scenario, with expected wages in legal professions rising by approximately 18% with the highest risk aversion scenario, compared to less than 16% in the baseline.

Figure A3 and Figure A4 show the results of the alternative scenario where the probabilities of automation and augmentation are set to $p_A = 0.8$ and $p_\gamma = 0.8$. The results are qualitatively similar to the baseline scenario, but the labor reallocation is stronger, since workers believe that AI is more likely to be eventually successful. As a result, training paths that are more exposed to automation (legal professions, business, and computer science) lose up to -25% of their workforce (compared to -16% in the baseline), while the safer paths (technical manual jobs, education, and no training) gain up to $+14\%$ (compared to $+10\%$ in the baseline). The wage impact is also stronger, with an increase for all the training paths of roughly $+20\%$ (compared to $+14\%$ in the baseline).

Finally, I test the sensitivity of the results to the Gumbel dispersion parameter ζ . In the baseline scenario, I set $\zeta = 0.053$, following Althoff and Reichardt (2026). As a robustness check, I also consider an alternative scenario with $\zeta = 0.3$, which implies that agents are more indifferent to differences in expected wages and training costs. The results can be seen in Figure A5 and Figure A6. Again, the results are qualitatively similar to the baseline scenario. Obviously, the labor reallocation is less pronounced, since the choices of agents are less sensitive to the differences in expected wages and training costs. The most negatively impacted paths (legal professions, business, and computer science) lose only up to -11% of their workforce (compared to -16% in the baseline). As a result, the wage impact is slightly more uniform. The total effect is still largely positive for all the training paths, with an increase of roughly 13% in the training paths most exposed to automation and an increase up to 17% in the least exposed training path (education).

Overall, the main implications of the model are robust to alternative parameterizations, confirming the importance of accounting for the labor supply response to AI. In all specifications, the qualitative results are the same: the most exposed paths (legal professions, business, and computer science) lose a significant share of their workforce, while the safer paths (technical manual jobs, education, and no training) gain a significant share of their workforce. This labor reallocation generates a general equilibrium scarcity effect that pushes up expected wages in the most exposed paths, while compressing wages in the safer paths. The net effect is a remarkably uniform wage increase across all training paths.

7 Conclusion

This paper studies how Artificial Intelligence reshapes human capital investment when its impact is uncertain and heterogeneous across training paths. I develop a task-based model in which risk-averse agents make irreversible training choices to maximize a certainty equivalent that penalizes AI-related wage variance, and I calibrate the framework to the US labor market using ACS and BLS data together with task-level automation and augmentation exposure from Althoff and Reichardt (2026) and Massenkoff and McCrory (2026).

The model delivers three main results. First, expected automation is the dominant driver of labor reallocation: workers systematically abandon highly exposed cognitive paths—legal professions, business, and computer science—whose employment shares fall by up to -16% , and crowd into technical manual jobs, education, and the no-training option. Expected augmentation, by contrast, plays a minor role in reallocation because its expected impact and uncertainty are relatively similar across training paths.

Second, the endogenous reallocation of labor generates an across-the-board increase in expected wages of approximately $+14\%$. The scarcity of workers in highly exposed paths pushes up the price of what they produce and their wages, while the congestion in safer paths compresses wages there. These two forces nearly cancel at the aggregate level, yielding a remarkably uniform wage increase that largely offsets the direct negative impact of automation on the most exposed training paths.

Third, automation uncertainty generates a pure general equilibrium risk premium of up to $+2.3\%$. Risk-averse individuals flee volatile training paths independently of their expected returns, creating additional labor scarcity that raises expected wages in those paths above what the expected productivity effects alone would predict. This channel is absent from models that treat AI as a deterministic shock, or that assume symmetric uncertainty across occupations.

Taken together, these results highlight the importance of accounting for the labor supply response to AI. A training path facing severe automation risk may nonetheless see rising expected wages due to the scarcity channel, while a path perceived as safe may experience wage compression from labor market congestion.

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A Robustness Figures

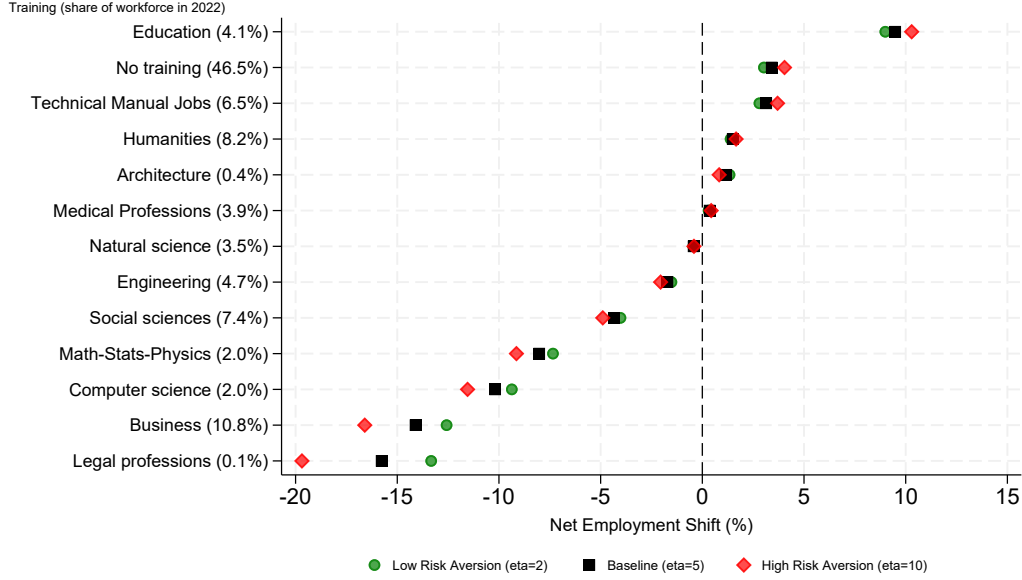


Figure A1: Employment Reallocation across Training Paths: Baseline vs Alternative Risk Aversion Scenarios.

Notes: This figure compares the net percentage change in employment shares (relative to the pre-AI 2022 baseline) across training paths under three risk aversion scenarios: low risk aversion ($\eta = 2$, green circles), baseline ($\eta = 5$, black squares), and high risk aversion ($\eta = 10$, red diamonds). All other parameters are held at baseline values ($p_A = p_\gamma = 0.5$, $\kappa = 0.5$, $\zeta = 0.053$). Higher risk aversion amplifies the flight from uncertainty (Proposition 2): risk-averse agents penalize uncertain training paths more heavily, leading to larger employment outflows from highly exposed paths (legal professions, business, computer science) and correspondingly larger inflows into safer paths (technical manual jobs, education, no training). However, the qualitative ranking of training paths is unchanged across all three scenarios. See Section 4 for details on the baseline calibration.

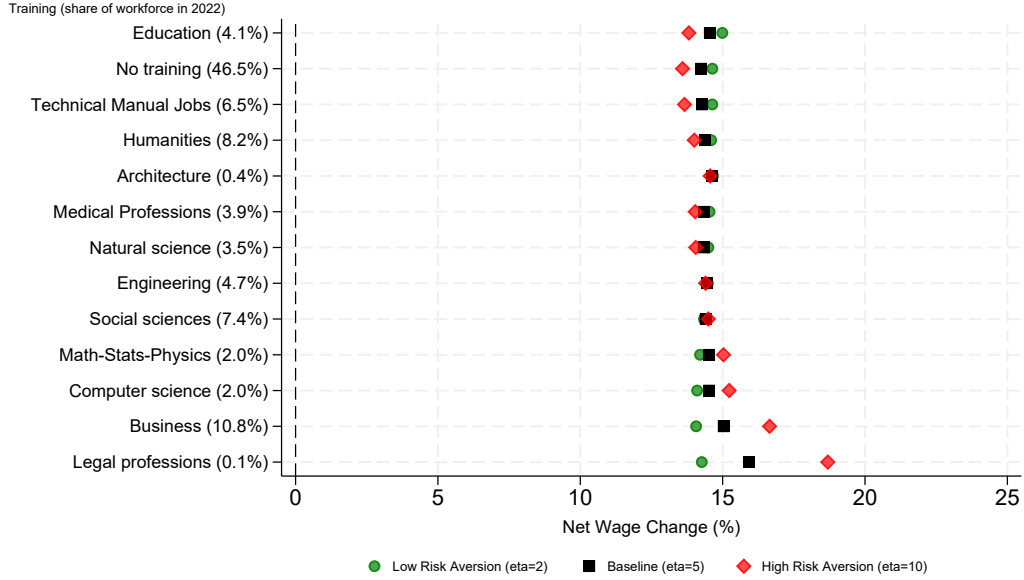


Figure A2: Wage Impact across Training Paths: Baseline vs Alternative Risk Aversion Scenarios.

Notes: This figure compares the net percentage change in expected log-wages (relative to the pre-AI 2022 baseline) across training paths under three risk aversion scenarios: low risk aversion ($\eta = 2$, green circles), baseline ($\eta = 5$, black squares), and high risk aversion ($\eta = 10$, red diamonds). All other parameters are held at baseline values ($p_A = p_\gamma = 0.5$, $\kappa = 0.5$, $\zeta = 0.053$). Higher risk aversion generates a larger endogenous risk premium: the amplified flight from uncertain paths creates additional labor scarcity that pushes up expected wages in those paths beyond the baseline prediction. For instance, legal professions see expected wages rise by approximately +18% under $\eta = 10$, compared to roughly +16% under $\eta = 5$. The wage increase remains positive for all training paths across all three scenarios. See Section 4 for details on the baseline calibration.

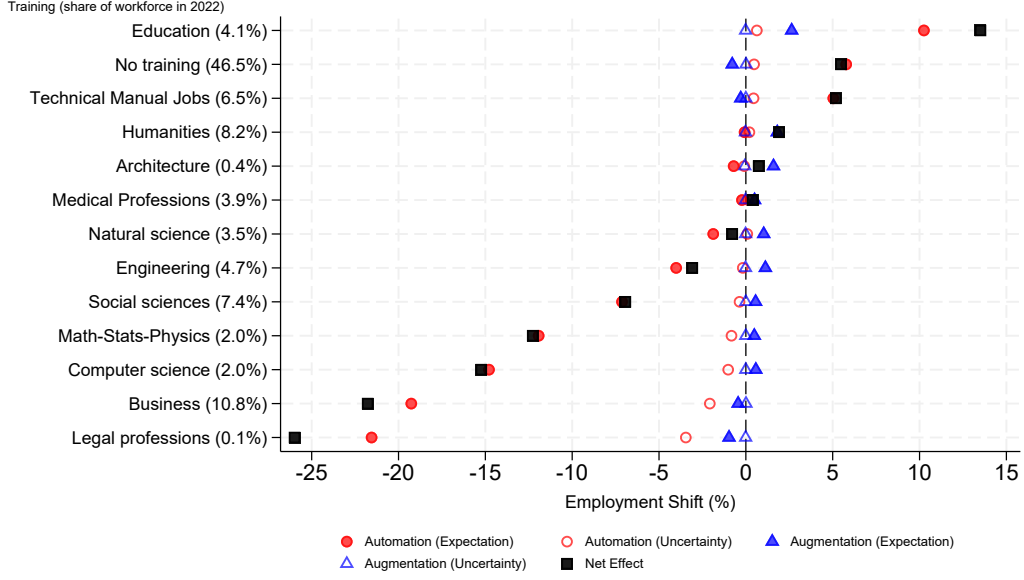


Figure A3: Employment Reallocation across Training Paths: Alternative Automation/Augmentation Probabilities ($p_A = p_\gamma = 0.8$)

Notes: This figure decomposes the percentage change in employment shares (relative to the pre-AI 2022 baseline) into four channels: expected automation (red circles), uncertainty over automation (red hollow circles), expected augmentation (blue triangles), and uncertainty over augmentation (blue hollow triangles). The total net effect is shown as black squares. This specification raises the adoption probabilities to $p_A = p_\gamma = 0.8$ (agents believe there is an 80% chance that theoretically possible automation or augmentation will materialize), with all other parameters at baseline values ($\eta = 5$, $\kappa = 0.5$, $\zeta = 0.053$). Compared to the baseline ($p_A = p_\gamma = 0.5$), the higher adoption probabilities amplify both expected shocks and their uncertainty, leading to stronger labor reallocation: the most exposed paths (legal professions, business, computer science) lose up to -25% of their workforce, while safer paths (technical manual jobs, education, no training) gain up to $+14\%$. The qualitative ranking of training paths is unchanged. See Section 3 for details on how the AI shocks are constructed.



Figure A4: Wage Impact across Training Paths: Alternative Automation/Augmentation Probabilities ($p_A = p_\gamma = 0.8$).

Notes: This figure decomposes the percentage change in expected log-wages into six components: expected automation with no labor reallocation (orange hollow diamond), expected automation with labor reallocation (red solid circle), expected augmentation with no labor reallocation (navy hollow diamond), expected augmentation with labor reallocation (blue solid triangle), uncertainty over automation (red hollow circles), and uncertainty over augmentation (blue hollow circles). The total net wage change is shown as black squares. This specification uses $p_A = p_\gamma = 0.8$, with all other parameters at baseline values ($\eta = 5$, $\kappa = 0.5$, $\zeta = 0.053$). Compared to the baseline ($p_A = p_\gamma = 0.5$), the higher adoption probabilities generate larger expected productivity gains and stronger labor reallocation, yielding an across-the-board wage increase of roughly +20% (compared to +14% in the baseline). The scarcity channel remains the dominant force: despite larger direct negative effects on highly exposed paths, the labor outflow from those paths pushes up their expected wages. See Section 3 for details on how the AI shocks are constructed.

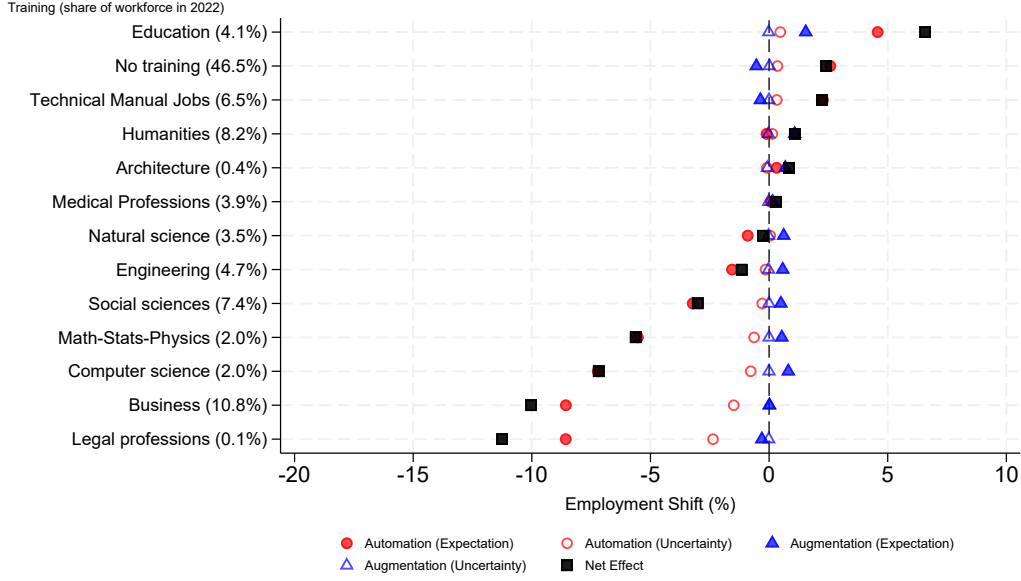


Figure A5: Employment Reallocation across Training Paths: Alternative Gumbel Dispersion Parameter ($\zeta = 0.3$).

Notes: This figure decomposes the percentage change in employment shares (relative to the pre-AI 2022 baseline) into four channels: expected automation (red circles), uncertainty over automation (red hollow circles), expected augmentation (blue triangles), and uncertainty over augmentation (blue hollow triangles). The total net effect is shown as black squares. This specification sets the Gumbel dispersion parameter to $\zeta = 0.3$ (compared to $\zeta = 0.053$ in the baseline), implying that agents' training choices are less sensitive to differences in expected wages and training costs. The structural costs C_j^* are re-calibrated to reproduce the observed 2022 employment shares under the new ζ . All other parameters are held at baseline values ($p_A = p_\gamma = 0.5$, $\kappa = 0.5$, $\eta = 5$). As expected, the higher dispersion dampens labor reallocation: the most exposed paths lose up to -11% (compared to -16% in the baseline). The qualitative ranking is unchanged—the same paths gain and lose workers—confirming that the reallocation patterns are not driven by an extreme sensitivity of the logit model. See Section 4 for details on the baseline calibration.

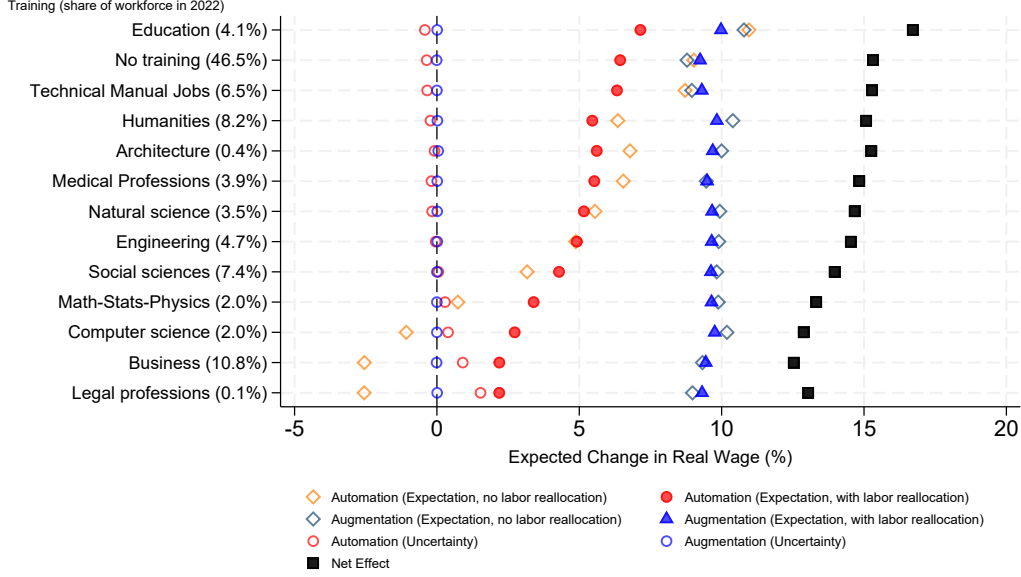


Figure A6: Wage Impact across Training Paths: Alternative Gumbel Dispersion Parameter ($\zeta = 0.3$).

Notes: This figure decomposes the percentage change in expected log-wages into six components: expected automation with no labor reallocation (orange hollow diamond), expected automation with labor reallocation (red solid circle), expected augmentation with no labor reallocation (navy hollow diamond), expected augmentation with labor reallocation (blue solid triangle), uncertainty over automation (red hollow circles), and uncertainty over augmentation (blue hollow circles). The total net wage change is shown as black squares. This specification sets the Gumbel dispersion parameter to $\zeta = 0.3$ (compared to $\zeta = 0.053$ in the baseline), with structural costs C_j^* re-calibrated accordingly. All other parameters are held at baseline values ($p_A = p_\gamma = 0.5$, $\kappa = 0.5$, $\eta = 5$). The dampened labor reallocation leads to a slightly more dispersed wage impact: training paths most exposed to automation see a total increase of roughly +13%, while the least exposed path (education) gains up to +17%. The “no labor reallocation” markers (hollow diamonds) are identical to those in the baseline, since the direct productivity effects do not depend on ζ ; only the reallocation-mediated wage effects differ. The net effect remains positive for all training paths. See Section 4 for details on the baseline calibration.